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ORIGINAL PAPER

Short-term water quality variable prediction using a hybrid CNN–LSTM deep learning model

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Abstract
Water quality monitoring is an important component of water resources management. In order to predict two water quality variables, namely dissolved oxygen (DO; mg/L) and chlorophyll-a (Chl-a; µg/L) in the Small Prespa Lake in Greece, two standalone deep learning (DL) models, the long short-term memory (LSTM) and convolutional neural network (CNN) models, along with their hybrid, the CNN–LSTM model, were developed. The main novelty of this study was to build a coupled CNN–LSTM model to predict water quality variables. Two traditional machine learning models, support-vector regression (SVR) and decision tree (DT), were also developed to compare with the DL models. Time series of the physicochemical water quality variables, specifically pH, oxidation–reduction potential (ORP; mV), water temperature (°C), electrical conductivity (EC; µS/cm), DO and Chl-a, were obtained using a sensor at 15-min intervals from June 1, 2012 to May 31, 2013 for model development. Lag times of up to one (t − 1) and two (t − 2) for input variables pH, ORP, water temperature, and EC were used to predict DO and Chl-a concentrations, respectively. Each model's performance in both training and testing phases was assessed using statistical metrics including the correlation coefficient (*r*), root mean square error (*RMSE*), mean absolute error (*MAE*), their normalized equivalents (*RRMSE*, *RMAE*; %), percentage of bias (PBIAS), Nash–Sutcliffe coefficient (E_{NS}), Willmott's Index, and graphical plots (Taylor diagram, box plot and spider diagram). Results showed that LSTM outperformed the CNN model for DO prediction, but the standalone DL models yielded similar performances for Chl-a prediction. Generally, the hybrid CNN–LSTM models outperformed the standalone models (LSTM, CNN, SVR and DT models) in predicting both DO and Chl-a. By integrating the LSTM and CNN models, the hybrid model successfully captured both the low and high levels of the water quality variables, particularly for the DO concentrations.

Keywords Long short-term memory · Convolutional neural network · Water quality modeling · Deep learning

1 Introduction

Water quality monitoring plays an important role in water resources management. It can provide decision makers and stakeholders with quantitative information to facilitate sustainable management of ongoing and emerging environmental problems (Wu and Liu 2012; Wu and Chen 2013). Specifically, water quality monitoring refers to the collection, measurement and analysis of water samples to determine the physicochemical and biological attributes of a water body (Sinshaw et al. 2019). Dissolved oxygen (DO) and chlorophyll-a (Chl-a) are two important water quality variables used to monitor lakes. The DO concentration of a water body indicates the equilibrium between the processes that produce oxygen (e.g., photosynthesis) and consume oxygen (e.g., chemical oxidation) (Fijani et al. 2019), and Chl-a provides an estimate of the concentration of algae in a water body, which is an indicator of eutrophication. Monitoring these water quality variables is required for the sustainable management of lake environments.

Conventional water quality monitoring techniques require manual collection of water samples to later be

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analyzed in a laboratory setting which is costly and time consuming (Wu and Liu 2012). Another conventional method involves the use of sensors, which have some drawbacks. The main disadvantage is that they measure few water quality variables, often with less accuracy than manual collection (Oelen et al. 2018). Predictive modeling using statistical and artificial intelligence approaches provide an alternative method for water quality monitoring. These models have many benefits: (1) reduce time and costs (e.g., material and labor costs), (2) facilitate prediction under various phases of a system, (3) simplify a complicated system to improve its understanding, and (4) predict target values when site access is problematic (Sinshaw et al. 2019). In recent decades, different predictive models have been successfully applied to predict water quality variables in water bodies, including the artificial neural network (Alizadeh and Kavianpour 2015; Barzegar et al. 2016a; Asadollahfardi et al. 2018; Sinshaw et al. 2019; Zhang et al. 2019; Goz et al. 2019; Zhu et al. 2019), fuzzy logic (Chen and Mynett 2003; Pereira et al. 2009; Huang et al. 2018), neuro fuzzy methods (Noori et al. 2013; Barzegar et al. 2016a; Khadr 2017; Aghel et al. 2019; Kisi et al. 2019; Zhu et al. 2019), M5P model-Tree (Yi et al. 2018), multivariate adaptive regression spline (MARS) (Kisi and Parmar 2016; Najafzadeh and Ghaemi 2019), extreme learning machines (ELM) (Fijani et al. 2019; Barzegar et al. 2018; Goz et al. 2019), random forest (Yajima and Derot 2018), decision tree (DT) (Jaloree et al. 2014) and least squares support vector machines (LSSVM) (Noori et al. 2015b; Kisi and Parmar 2016; Fijani et al. 2019; Najafzadeh and Ghaemi 2019). However, the complexity and non-linearity of water quality variables make it challenging for these models to recognize the processes that affect water quality. Therefore, more advanced prediction models are still required to improve the accuracy of water quality variable prediction.

Deep learning (DL), an advanced sub-field of machine learning (ML) for artificial intelligence, has been increasing in popularity recently. It has been successfully applied to many fields of science including computer vision (Fang et al. 2019), image classification (Affonso et al. 2017; Liu et al. 2019c; Fan et al. 2019), speech recognition (Fayek et al. 2017; Cummins et al. 2018), time series and spatiotemporal prediction (Chen et al. 2018; Yang and Chen 2019; Xu et al. 2019; Bui et al. 2019) and natural language processing (Shin et al. 2017; Plappert et al. 2018). DL incorporates several ML algorithms that extract high-level representations from complex data through a hierarchical learning process using structures consisting of multiple nonlinear transformations (Zuo et al. 2019).

Long short-term memory (LSTM) and convolutional neural networks (CNN) are two popular DL models. The LSTM is a type of recurrent neural network (RNN) which collects extended sequential data in the hidden memory for processing, representation and storage. Also, it updates over time to ensure constancy in the time information (Kim and Cho 2019). The CNN is composed of a sequence of convolutional layers, where each neuron in a layer is connected to a small local region of neurons in the input data. This is executed by sliding a weight matrix, called a filter, over the input and the convolution (or dot product) computed at each point, which is referred to as the feature map, between the input and the filter. This architecture enables the model to learn the filter needed to recognize identified patterns in the input data (Borovykh et al. 2017). Research on the application of DL models to the field of hydrology has recently been growing. For example, Tao et al. (2016) used a stacked denoising autoencoder with four layers and 1000 hidden nodes to extract precipitation information from 15×15 pixel satellite cloud images. Song et al. (2016) used a deep belief network with observed land surface temperature and leaf area index as inputs to predict field-measured soil moisture. Zhang et al. (2018) built different neural network models, including LSTM and a gated recurrent network (GRU), to predict the water level of a combined sewage outflow structure. Kratzert et al. (2018) used LSTM to model the rainfall-runoff process of 241 catchments. More specifically, certain DL methods have been used to predict water quality variables. For example, Liu et al. (2019b) used an LSTM deep neural network in an Internet of Things (IoT) environment to predict drinking-water quality in Yangzhou, China. Chen et al. (2019) proposed a multivariate deep belief network model with particle swarm optimization to predict ammonia–nitrogen concentrations in aquaculture ponds. Recently, Chl-a concentration in water bodies has been predicted using DL models. Choi et al. (2019) applied a CNN model to predict Chl-a concentration in Daechung Lake, South Korea, with emphasis on handling data imbalance and skewness. They indicated that using log transformation and oversampling techniques together helped improve the performance of the CNN-based prediction model, especially for small regions. Cho et al. (2018) used LSTM for multi-step ahead prediction of Chl-a in Gongju, South Korea. Their results showed that the LSTM network model achieved higher accuracy than the dense network model, and batch normalization as a regularization method helped the learning process.

Hybrid neural networks, that incorporate the advantages of several networks, have gained increasing attention in recent years. It has been found that the combination of CNN and LSTM can be useful to develop accurate predictive models (Kim and Cho 2019). Huang et al. (2018) found that it is advantageous to perform feature extraction using CNN, and then input the feature values in the LSTM architecture. In the hybrid CNN–LSTM model, the one

dimensional-CNN extracts deep features of the main elements. Then, the LSTM model is applied to perform the prediction by using these in-depth features. A combined CNN–LSTM model was applied to predict crash risk (Li et al. 2020), residential energy consumption (Kim and Cho 2019), as well as household power consumption (Kim and Cho 2018, 2019).

The main focus of this research is to: (1) explore the capability of DL models, namely the LSTM, CNN and hybrid CNN–LSTM models, compared to traditional ML models (the support vector regression (SVR) and decision tree (DT) models) in predicting water quality variables (DO and Chl-a concentrations); (2) compare the performances of the LSTM and CNN methods for short-term water quality variable prediction; and (3) develop a hybrid model (CNN–LSTM) that combines the advantages of the CNN and LSTM methods. Although, as mentioned above, previous studies explored the ability of these methods to predict water quality variables, no studies have compared their performances. Also, to the best of the authors' knowledge, this is the first time a coupled CNN–LSTM model has been used to predict water quality variables.

2 Materials and methods

2.1 Case study area and data collection

The Prespa basin is situated in the Balkan peninsula in southeastern Europe and is shared between three countries: Albania, Greece and Macedonia. The basin is comprised of two inter-linked lakes, Big Prespa and Small Prespa. These lakes are separated by an alluvial isthmus situated in Greece (Panagiotopoulos et al. 2013) and the lakes are connected by a small artificial channel (Krstić 2012). Small Prespa Lake has a total area of 47.4 km^2, of which 43.5 km^2 belong to Greece and 3.9 km^2 to Albania (Fig. 1). This study focuses on predicting water quality variables solely for Small Prespa Lake. The lake is mostly recharged by runoff from different surface bodies and lateral sub-surface flow that occurs due to cross flows in the interacting aquifer systems of the area (Tziritis 2014). Another important source of recharge for the lake, during specific periods of the year, comes from the Devoll River into the southwest portion of the lake, on the Albanian side (Fijani et al. 2019). The Prespa area experiences a hot and dry climate during the summer, with a mean annual temperature of 23.6 °C in July. In contrast, the winters are cold with a mean temperature of 0.8 °C in January (Hollis and Stevenson 1997; Fijani et al. 2019). The average annual rainfall and air temperature are 647 mm and 14 °C, respectively.

In recent decades, anthropogenic pressures on Small Prespa Lake have significantly decreased the water level and caused deterioration of the water quality. The water loss is mainly due to intensified irrigated agricultural practices (Patceva and Mitic 2010). Moreover, the lake may be threatened by eutrophication due to elevated concentrations of Chl-a. Therefore, monitoring the water quality in this lake is increasingly important. The water quality data used to develop the models in this study were gathered using sensors between June 1, 2012 and May 31, 2013 at 15-min intervals (Tziritis 2014). The physico-chemical variables, including electrical conductivity (EC), oxidation–reduction potential (ORP), pH, water temperature, dissolved oxygen (DO) and chlorophyll-a (Chl-a), were measured by a multi-probe sensor CYCLOPS-7 of TURNER DESIGS® in Small Prespa Lake. The sensor was deployed at a depth of 1.5 m at the northern side of the lake. The sensor reached a total depth of approximately 3 m, with minor fluctuations throughout the year (Tziritis 2014).

Descriptive statistics of the water quality data used in this study, including minimum, maximum, mean, standard deviation, skewness and kurtosis, can be found in Table 1. Standard deviation was highest for the EC (57.33 µS/cm) and ORP (52.63 mV), while pH (0.19) had the lowest standard deviation. The DO concentration ranged between 0.02 and 14.97 mg/L, with a mean of 8.83 mg/L. The minimum and maximum values of Chl-a were 0.6 and 22.8 µg/L (with a mean of 2.67 µg/L). The skewness and kurtosis for the ORP (− 1.64 and 2.98 mV, respectively) and Chl-a (2.65 and 14.76 µg/L, respectively) variables were outside the range of − 2 and + 2, indicating that these variables did not follow a normal (Gaussian) distribution (Barzegar et al. 2016b).

2.2 Long short-term memory (LSTM)

LSTM neural networks are an improvement over RNNs, which were designed for sequence prediction, because they address the stability restriction (i.e., the vanishing gradient problem) of traditional RNNs by combining gating functions with state dynamics (Hochreiter and Schmidhuber 1997). An LSTM network includes several memory blocks that are connected via layers. Each layer is comprised of a set of recurrently connected memory cells and three multiplicative units: the input, forget and output gates (Yuan et al. 2018). The input gate provides the information to the cell state in three steps. First, input values added to the cell state are regulated with the sigmoid function. Second, a vector is created containing all possible values to be added to the cell state using the hyperbolic tangent function. Finally, the regulatory filter is multiplied by the newly created vector, and the information is added to the cell

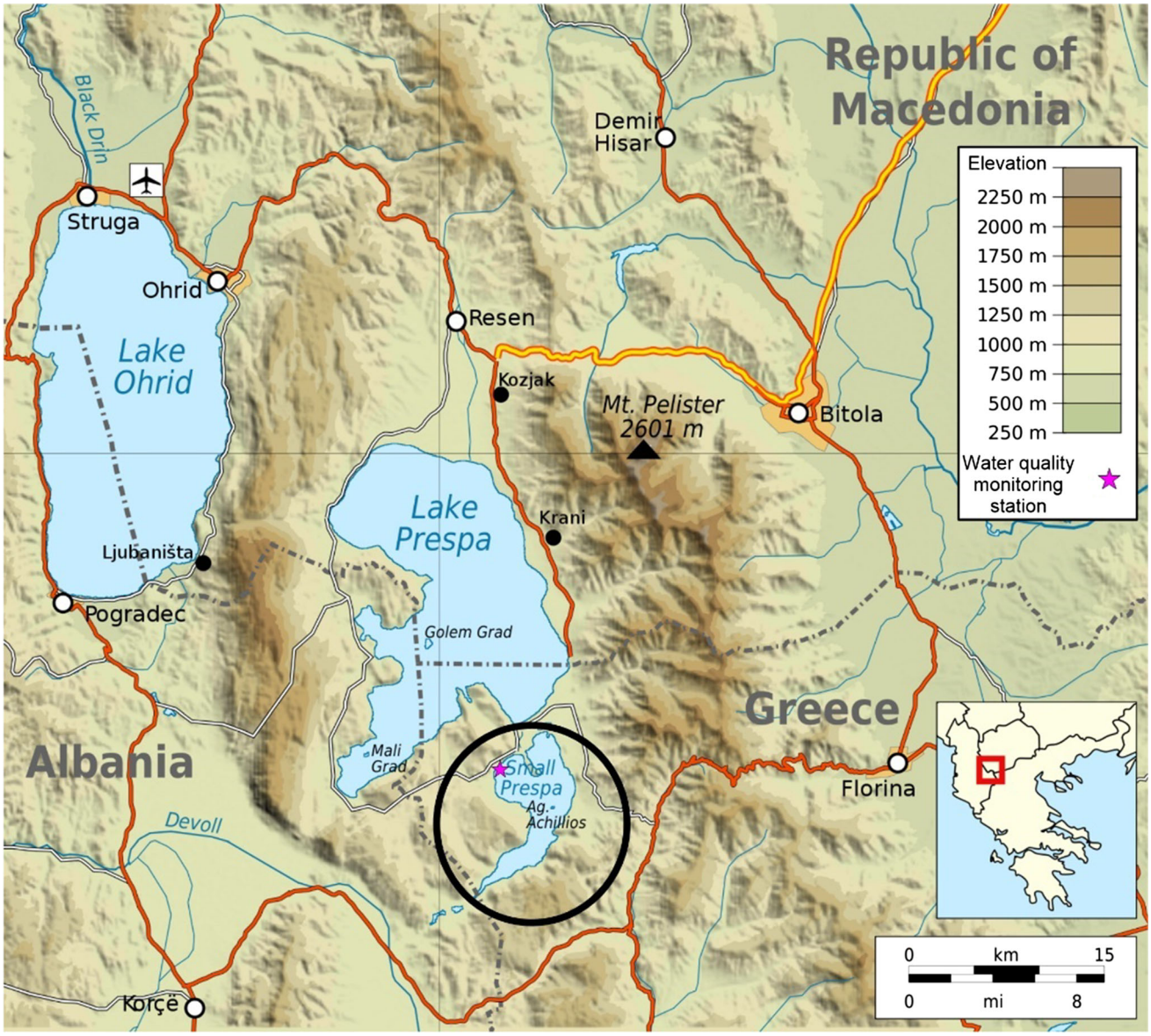


Fig. 1 Location of Small Prespa Lake. Adapted from https://commons.wikimedia.org/wi

Table 1 Statistical summary of the physicochemical water quality variables in Small Prespa Lake

Water quality variable	Min	Max	Mean	SD	Skewness	Kurtosis
EC (μS/cm)	247.00	493.00	325.59	57.33	0.87	− 0.23
ORP (mV)	139.10	474.30	392.29	52.63	− 1.64	2.98
pH	6.80	8.60	8.04	0.19	− 0.22	0.41
Water temperature (°C)	2.90	29.00	15.48	8.14	− 0.03	− 1.49
DO (mg/L)	0.02	14.97	8.83	2.26	0.09	− 1.15
Chl-a (μg/L)	0.60	22.80	2.67	1.50	2.65	14.76

state. Using a multiplying filter, the forget gate removes the information that is no longer required for the LSTM to understand processes or the information of lesser importance. The output gate chooses the useful information from the current cell state and displays it as the output.

The architecture of the LSTM model and the structure of an LSTM cell are shown in Fig. 2. Suppose the input time series data is $x = (x_1, x_2, \ldots, x_{t-1}, x_t)$. A typical LSTM cell contains sigmoid (σ) and hyperbolic tangent layers, along with pointwise summation ($\oplus$), and multiplication ($\otimes$)

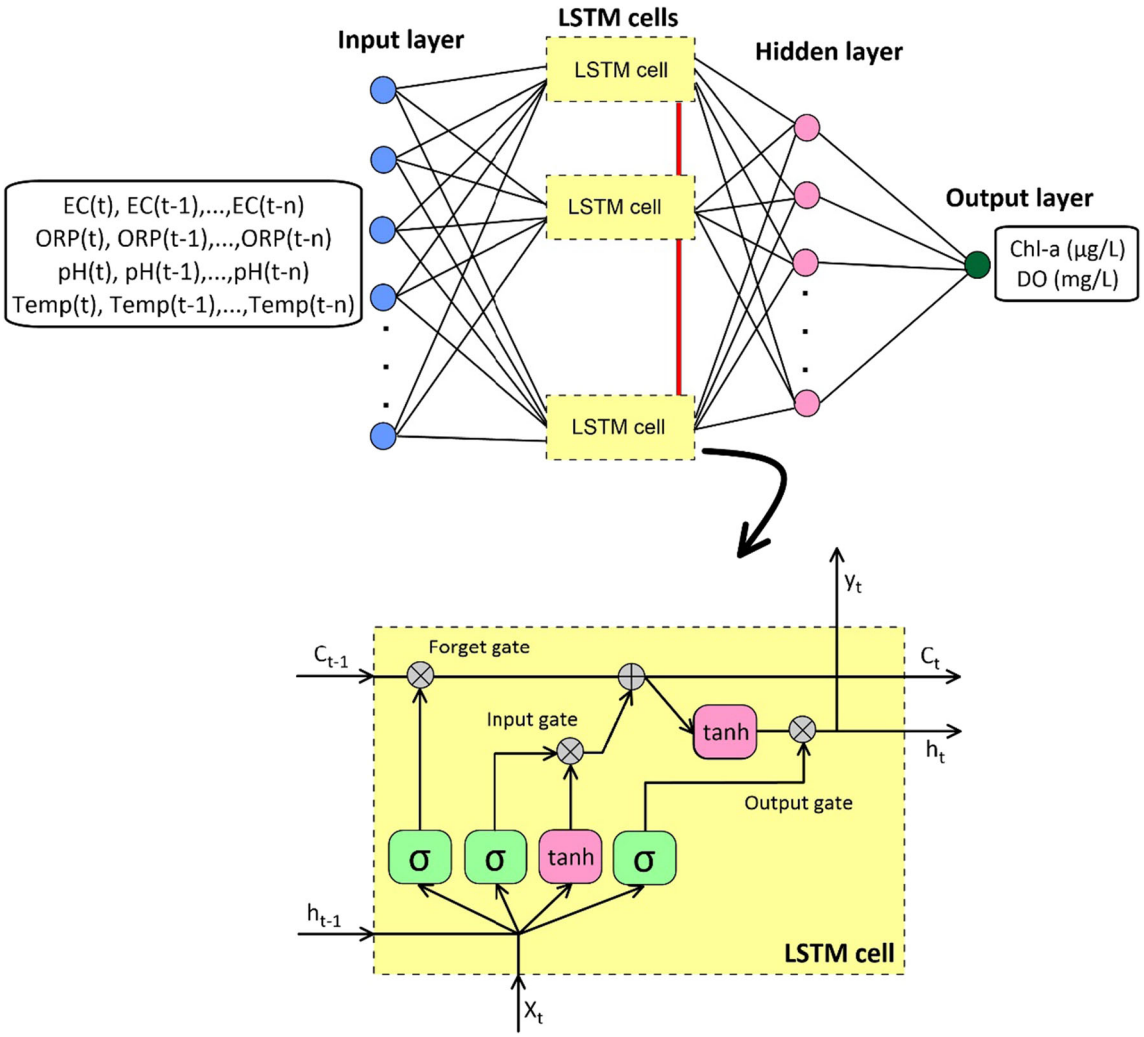


Fig. 2 Architecture of the LSTM model and LSTM cell structure (Wu and Lin 2019)

operations. The LSTM can predict the target variables $y = (y_1, y_2, \ldots, y_{t-1}, y_t)$ by updating the gates (input gate i_t, forget gate f_t and output gate y_t) on the memory cell c_t, from time $t = 1$ to T. The mathematical equations for the LSTM are defined as follows (Wu and Lin 2019):

$$i_t = \sigma(w_i x_t + R_i h_{t-1} + b_i) \quad (1)$$

$$f_t = \sigma(w_f x_t + R_f h_{t-1} + b_f) \quad (2)$$

$$y_t = \sigma(w_y x_t + R_y h_{t-1} + b_y) \quad (3)$$

$$c_t = f_t c_{t-1} + i_t \bar{c}_t \quad (4)$$

$$\bar{c}_t = \sigma(w_c x_t + R_c h_{t-1} + b_c) \quad (5)$$

$$h_t = \mathrm{y}_t \sigma(c_t) \quad (6)$$

where x_t is the input vector, w_i, w_f, and w_y denote the matrix of weights from the input, forget, and output gates to the input, respectively, R_i, R_f, and R_y define the matrix of weights from the input, forget, and output gates to the input, respectively, b_i, b_f, b_y are the input, forget, and output gate bias vectors, respectively, c_{t-1} and h_{t-1} are the previous cell and its output vector, respectively, and h_t is the output vector.

2.3 Convolutional neural network (CNN)

The CNN model consists of three layers: the input, hidden and output layers. The input of a three-dimensional array is usually fed to a convolutional layer where the dimensions are represented by height, weight and the number of channels. Assuming a one-dimensional input $x = (x_t)_{t=0}^{N-1}$ of size N with no zero padding in the first layer, the feature output map is generated through convolving this input with a set of M_1 three-dimensional filters, w_h^1 for $h = 1, \ldots, M_1$, in which the filters are employed over all the input channels (Borovykh et al. 2017):

$$a^1(i,h) = \left(w_h^1 \times x\right)(i) = \sum_{j=-\infty}^{\infty} w_h^1(j) \times (i-j) \quad (7)$$

where $w_h^1 \in R^{1\times k\times 1}$ and $a^1 \in R^{1\times N-k+1\times M_1}$. There will be one input channel, and the output of the first layer is then passed through the non-linear activation function $h(\cdot)$ to give $f^1 = h(a^1)$.

The hidden layer includes a convolutional layer, pooling layer, and fully connected layer. The convolutional layer is responsible for automatically extracting features at various regions of the entire raw input or the intermediate feature maps with learnable filters (Zuo et al. 2019). Neurons in one convolution layer are not connected to all the neurons in the next layer but only to a small region of it. Therefore, the filter employs a shared weight matrix to perform the convolutional operation. Note that the weights are updated during the training process (Hoseinzade and Haratizadeh 2019). The pooling layer converts all the values in the pooling window to one value. This transformation decreases the size of the input layer with a max-pooling operation, which selects the maximum value from each subarea of the previous layer (Zuo et al. 2019). Moreover, this layer reduces the computational cost of the learning process and handles any overfitting issues (Hoseinzade and Haratizadeh 2019).

In the hidden layer $l = 2, \ldots, L$, the input feature map $f^{l-1} \in R^{1\times N_{l-1}\times M_{l-1}}$, where $1 \times N_{l-1} \times M_{l-1}$ is the size of the output filter map from the previous convolution with $N_{l-1} = N_{l-2} - k + 1$, is convolved with a set of M_1 filters $w_h^1 \in R^{1\times k\times M_{l-1}}$, $h = 1, \ldots, M_1$, to create a feature map $a^1 \in R^{1\times N_l\times M_l}$ as follows (Borovykh et al. 2017):

$$a^1(i,h) = \left(w_h^l \times f^{l-1}\right)(i) = \sum_{j=-\infty}^{\infty}\sum_{m=1}^{M_{l-1}} w_h^l(j,m) f^{l-1}(i-j,m) \tag{8}$$

The fully connected layer flattens the high-level extracted features that are learned by the convolution layer, and incorporates them to obtain the final output. Then the resulting feature values are passed through the non-linear activation functions to give $f^1 = h(a^1)$. After L convolutional layers, the output of the network will be the matrix f^L, whose size depends on the filter size and number of filters used in the final layer (Borovykh et al. 2017). Figure 3 represents the architecture of the CNN model.

2.3.1 CNN–LSTM hybrid model

Both CNN and LSTM models have their specific features. In this study, a hybrid CNN–LSTM DL model, which integrates the advantages of both the CNN and LSTM models, was developed to predict water quality variables. First, the upper layer of the CNN–LSTM architecture was composed of CNN layers. CNN includes an input layer that enters the input variables, an output layer that extracts features to LSTM cells, and several hidden layers. The hidden layer typically includes a convolution layer, an activation function, and a pooling layer. The output of the LSTM is used as the input to the fully connected layer. The CNN layers are used to learn sequential features of the inputs, which conventional neural networks cannot do, and the lower LSTM layer incorporates these features by handling long-range dependencies for prediction of the target values. The topological architecture of the CNN–LSTM model is presented in Fig. 4.

2.4 Support vector regression (SVR)

SVR is a classical supervised machine learning method that was first proposed by Cortes and Vapnik (1995). Initially, the purpose of the SVR was to convert a low-dimensional nonlinear feature into a high-dimensional linear feature through kernel functions to avoid dimensionality (Huang et al. 2019). It is implemented through the frame of statistical learning theory and the minimal structure risk principle (Lei et al. 2019). Many papers and books (Vapnik 1995; Cortes and Vapnik 1995; Li et al. 2013; Noori et al. 2009, 2011, 2015a; Barzegar et al. 2017; Yu et al. 2017; Babaei et al. 2019) describe the principles and mathematical equations entailed in SVR; therefore, this paper does not include this.

2.5 Decision tree (DT)

DT is a non-parametric white-box supervised ML method (Quinlan 1986) based on a collection of decision rules (i.e., if–then rules). This method is composed of three types of nodes: (1) root nodes, also called decision nodes, which represent a choice that brings the subdivision of all samples into two or more similarly independent subsets, (2) internal nodes, also referred to as chance nodes, which indicate the possible choice of the feature (or attribute) in the tree, and (3) leaf nodes, also termed end nodes, which show the outcome as an aggregation of decisions (Song and Ying 2015). The model is created top-down in the form of a tree where the training data are partitioned in a recursive manner from a root node, followed by subsequent splitting points (internal nodes), which end in distinct leaves where the division terminates (Bacal et al. 2019). Each node in a decision tree is designated by an attribute which is chosen according to its ability to decrease chaos in the classification (Bacal et al. 2019). Additional details about the DT model can be found in Quinlan (1986) and Song and Ying (2015).

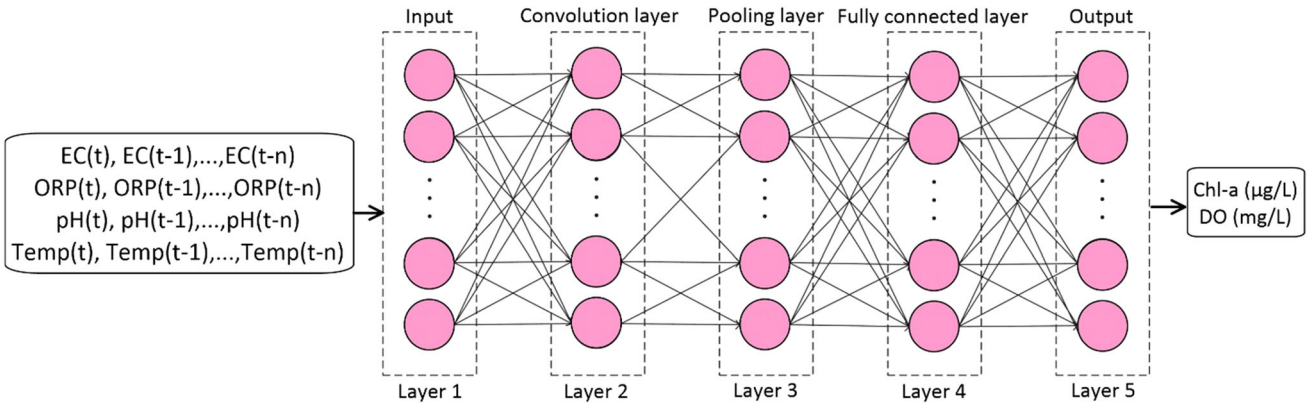


Fig. 3 The structure of the CNN model (Zuo et al. 2019)

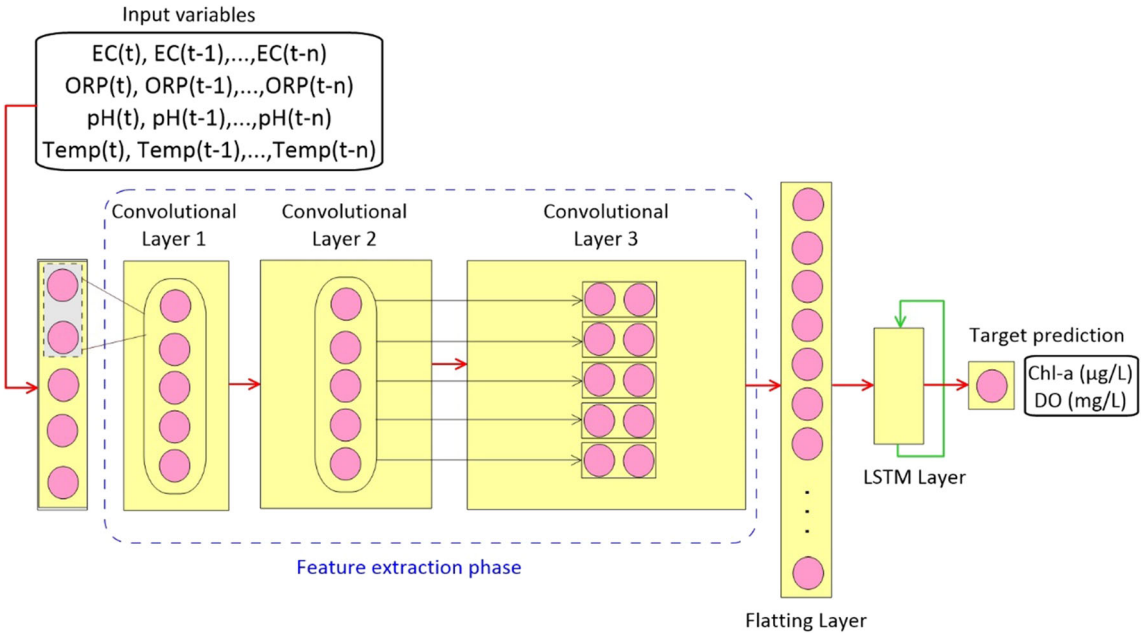


Fig. 4 The structure of the hybrid CNN–LSTM model (Liu et al. 2019a)

2.6 Model performance

The performance of the different models was evaluated using the correlation coefficient (r), root mean square error (*RMSE*), mean absolute error (*MAE*), and their normalized equivalents expressed as a percentage (*RRMSE, RMAE*), percentage of bias (*PBIAS*), Nash–Sutcliffe coefficient (E_{NS}) and Willmott's Index (*WI*). Equations and explanations of all these statistical metrics are widely discussed elsewhere (Ghorbani et al. 2018; Barzegar et al. 2019; Fijani et al. 2019).

The r metric represents the strength of the linear regression between observed and predicted values. The strongest linear relationship occurs when r equals 1. The *RMSE* and *MAE* express the accuracy of the models. Both metrics can range from 0 to ∞, where 0 indicates the highest accuracy. Model precision was evaluated with *RRMSE* and *RMAE*, which compare the percentage deviation from the predicted data. The performance of the model for *PBIAS* is considered very good, good, satisfactory or unsatisfactory if $PBIAS < \pm 10$, $\pm 10 \leq PBIAS < \pm 15$, $\pm 15 \leq PBIAS < \pm 25$ or $PBIAS > \pm 25$, respectively (Legates and McCabe 1999). According to the E_{NS} metric, each model's performance is categorized as very good, good, satisfactory, acceptable or unsatisfactory if $0.75 < E_{NS} \leq 1.00$, $0.65 < E_{NS} \leq 0.75$, $0.50 < E_{NS} \leq 0.65$, $0.40 < E_{NS} \leq 0.50$ or $E_{NS} \leq 0.4$, respectively (Moriasi et al. 2007; Khosravi et al. 2018). The *WI* metric is used to detect sensitivity to outliers in the observed data or insensitivity to the additive or proportional differences between predictions and observations,

which can occur with *r*, *RMSE*, *MAE* and *BIAS* metrics (Willmott 1981; Yaseen et al. 2018).

2.7 Predictive model implementation

Easily measured and accessible physicochemical water variables (e.g., pH, ORP (mV)), water temperature (°C) and EC (μS/cm) were used to develop the predictive models for water quality variables (i.e., DO (mg/L) and Chl-a (μg/L)) in Small Prespa Lake, Greece. First, the dataset of each prediction model was partitioned into two subsets: 70% for the training stage and 30% for the testing stage. The training dataset was used to develop the models, while the testing dataset was used to validate and compare the performance of models developed in the training phase. All input and output variables of the models were scaled between 0 and 1 via normalization and minimum–maximum scaling techniques, executed in the scikit-learn preprocessing library (Pedregosa et al. 2011) using Python. This process prevents dramatic changes in gradient and smooths the convergence. Also, all training and testing datasets were transformed into a supervised learning frame with the 'series_to_supervised' Python function, in which time series sequences are converted to input–output pairs using the sliding windows procedure (Cai et al. 2019). This was executed using lag observations (i.e., $t-1$) as inputs and the current observation (t) as the output variable. The best lag times were determined by trial and error and indicated by the lowest *RMSE*. Therefore, lag times of up to one ($t-1$) and two ($t-2$) for the input variables, which included pH, ORP, water temperature and EC, were used when developing the DO and Chl-a prediction models, respectively.

The LSTM models consist of an input layer, where the inputs (i.e., pH, ORP (mV), water temperature (°C) and EC (μS/cm)) and their lag times were imported to all three layers. To develop this model, an LSTM layer with an activation function represented by an Exponential Linear Unit (ELU), and having units of 64 and 32 for Chl-a and DO predictions, was used in the hidden layer. To avoid overfitting, a dropout layer with a rate of 0.001 was applied after the LSTM layer. A fully connected layer, termed "dense", with a unit of 1 and a linear activation function, were used. Then, the model was compiled with a mean squared error (*MSE*) loss function and an adaptive gradient algorithm (AdaGrad) optimizer with a learning rate of 0.01. Figure 5 shows the loss function plots for the models.

The CNN model for Chl-a prediction was trained with a hidden layer, where a convolution layer (Conv1D) with a filter size of 128, a kernel size of 2, the same padding type, an ELU activation type, and a uniform kernel initializer were used. Also, a pooling layer with max-pooling and a pooling size of 1, was applied where the maximum value from each neuron cluster was in the previous layer. A dropout layer with a rate of 0.001 was used after the pooling layer in the CNN to adjust for overfitting; the pooling layer can also be helpful to overcome overfitting (Kim and Cho 2019). A fully connected layer, which is called "dense", was applied to the output layer where a linear activation function was used. Finally, the model was compiled with an *MSE* loss function and an AdaGrad optimizer with a learning rate of 0.01. For the DO prediction, the CNN model had a similar structure except that a filter size of 32 was used. The loss plots for the CNN models are presented in Fig. 5.

In the hybrid model, the convolutional layer, the hidden layer and the pooling layer, which are part of the CNN model, were formatted in the same way as those of the standalone CNN model. Similarly, the LSTM part was formatted like the standalone LSTM model. The structures of the DL models are listed in Table 2.

The SVR models were constructed using the epsilon-SVR kernel-type equation. Different kernel function types, e.g., linear, radial basis function (RBF), sigmoid, and polynomial, were considered when developing the SVR models. The RBF was the optimal kernel, with the smallest error for both DO and Chl-a prediction. It performed better than other kernel functions because it has fewer hyper-parameters, which decrease the complexity of the model. While training the SVR model with the RBF kernel function, the regulation factor (C), epsilon (ε), and gamma (γ) hyper-parameters were set. A grid search with tenfold cross-validation, as suggested by Noori et al. (2009, 2011), was employed in the training phase for hyper-parameter tuning. In this procedure, samples were chosen from the space of the independent variables. Then, an error score was computed for each sample prediction and compared with the error scores found from the previous iterations. If the obtained error score at an iteration was lower than the previous one, the predictions with the lower error score were retained. This process was repeated until the end of iterations was reached. For the DO model, the optimal values of the C, ε, and γ were 1, 0.01, and 0.1, respectively, while the optimal values for the Chl-a model were 1, 0.1, and 0.1, respectively. The scikit-learn library (Pedregosa et al. 2011) for Python was used to implement the SVR models.

When developing the DT models, the maximum depth parameter had to be selected. It was chosen by trial and error using different maximum depth values between 1 and 10 at intervals of 1. The optimal maximum depths, based on the lowest *RMSE,* for the DO and Chl-a prediction models were 8 and 3, respectively.

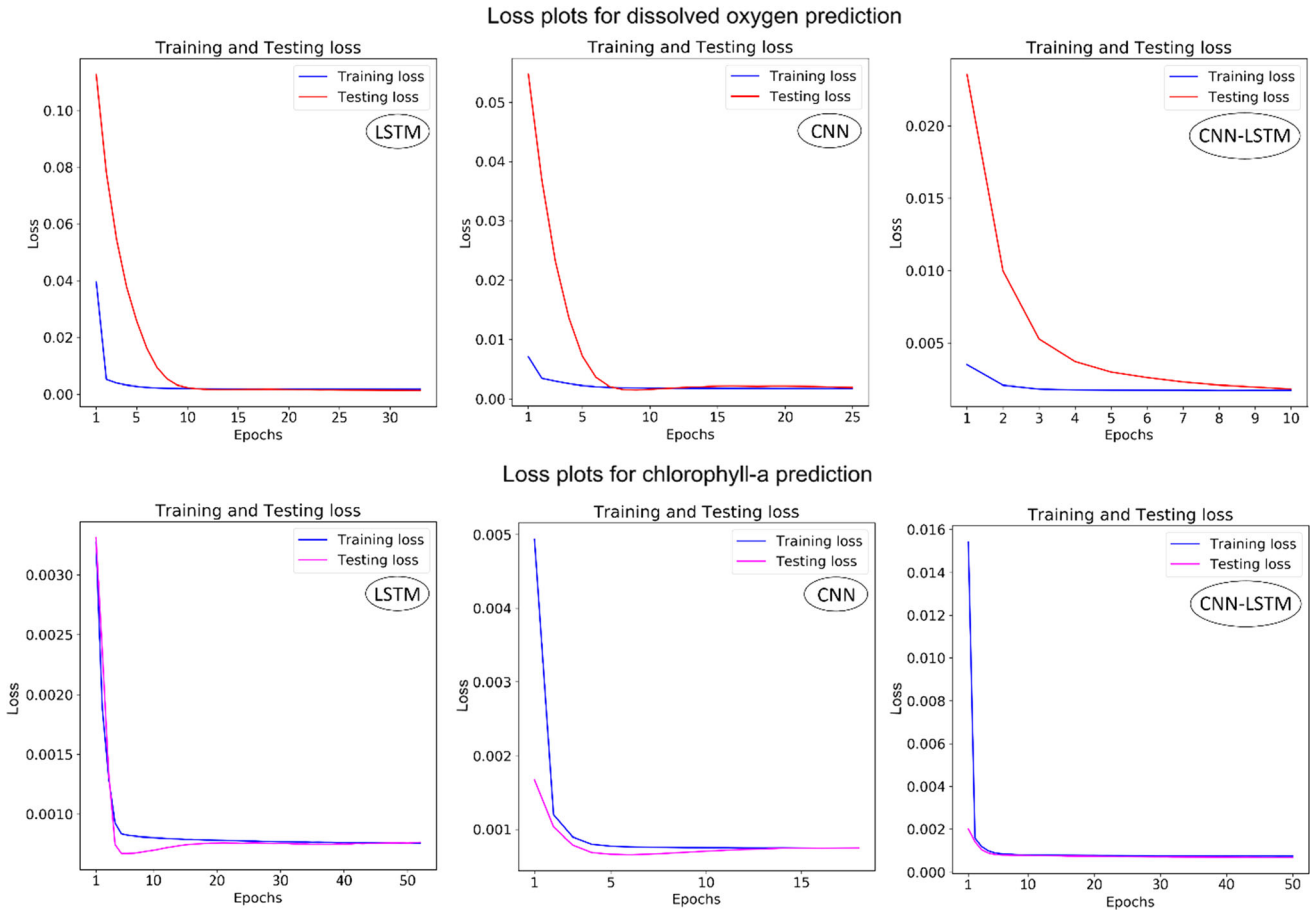


Fig. 5 Loss function plots for the DL models developed to predict DO and Chl-a

3 Results and discussion

The objective of this research was to investigate the performance of three DL models, namely the LSTM, CNN, and hybrid CNN–LSTM models, to predict water quality variables (i.e., DO and Chl-a). Also, two traditional ML models, the SVR and DT models, were developed to compare their performances to that of the DL models. After training the abovementioned models, the testing data were fed into the models to predict the target variables. The models were evaluated and compared using statistical indicators, including *r*, *RMSE*, *RRMSE*, *MAE*, *RMAE*, *PBIAS*, *ENS* and *WI,* in the training and testing periods. The results are summarized in Table 3.

For DO prediction, the LSTM model ($r = 0.968$, $RMSE = 0.545$mg/L, $MAE = 0.410$mg/L, $E_{NS} = 0.932$ and $WI = 0.984$) outperformed the CNN model ($r = 0.967$, $RMSE = 0.658$mg/L, $MAE = 0.522$ mg/L, $E_{NS} = 0.902$ and $WI = 0.977$) in the training and testing phases. This is consistent with results from a previous study by Gu et al. (2019), which concluded that the LSTM model performed better than the CNN model in time series prediction. This is likely due to the fact that the CNN model cannot capture the long-term dependency. The hybrid CNN–LSTM ($r = 0.970$, $RMSE = 0.518$mg/L, $MAE = 0.392$mg/L, $E_{NS} = 0.939$ and $WI = 0.982$) model outperformed the standalone models in the testing period for DO prediction. Overall, the *RMAE* of the CNN–LSTM-based DO prediction model was 0.24% and 1.45% lower than the LSTM and CNN models in the testing period, respectively. Figure 6 illustrates the observed and predicted DO values computed by the models in the testing period. Interestingly, in the testing period, the LSTM model performed better when predicting higher levels of DO, while the CNN model was better at predicting lower levels of DO. The hybrid CNN–LSTM model better captured both the lower and higher levels of DO concentrations by combining the strengths of the standalone models. All established DL models outperformed the traditional ML models (i.e., SVR and DT models) when predicting DO in the testing period. The CNN–LSTM model had an *RMAE* 2.55% and 1.18% lower than the SVR and DT models, respectively. Based on the *PBIAS* and E_{NS} values, all had good performances in predicting DO in the testing period.

Table 2 The structures of the developed DL models

Model	Predicted variable	Structure
LSTM	DO	1 LSTM layer (32 neurons, ELU activation) + 1 Dropout layer (0.001) + 1 Dense layer (1 neuron, linear activation) + Compile (MSE loss, Adam optimizer with a learning rate of 0.01)
	Chl-a	1 LSTM layer (64 neurons + ELU activation) + 1 Dropout layer (0.001) + 1 Dense layer (1 neuron, linear activation) + Compile (MSE loss, Adam optimizer with a learning rate of 0.001)
CNN	DO	Convolutional layer (32 filters + 2 filter sizes + ELU activation + padding 'same' + 1 stride) + Maxpooling (1 pooling size) + Flatten + Dense layer (12 neurons, linear activation) + 1 Dropout layer (0.001) + 1 Dense layer (1 neuron) + Compile (MSE loss, Adagrad optimizer with a learning rate of 0.01)
	Chl-a	Convolutional layer (128 filters + 2 filter sizes + ELU activation + padding 'same' + 1 stride) + Maxpooling (1 pooling size) + Flatten + Dense layer (30 neurons, SELU activation) + 1 Dropout layer (0.001) + 1 Dense layer (1 neuron) + Compile (MSE loss, Adagrad optimizer with a learning rate of 0.002)
CNN–LSTM	DO	Dense layer (32 neuros) + Convolutional layer (32 filters + 1 filter size + ELU activation + padding 'same' + 1 stride) + Maxpooling (1 pooling size) + 1 Dropout layer (0.001) + Convolutional layer (64 filters + 2 filter sizes + ELU activation + padding 'same' + 1 stride) + MaxPooling (2 pooling size) + 1 Dropout layer (0.001) + 1 LSTM layer (64 neurons + ELU activation) + 1 Dropout layer (0.001) + Flatten + 1 Dense layer (1 neuron, linear activation) + 1 Dropout layer (0.001) + 1 Dense layer (1 neuron) + Compile (MSE loss, Adagrad optimizer with a learning rate of 0.01)
	Chl-a	Dense layer (256 neuros) + Convolutional layer (128 filters + 1 filter size + ELU activation + padding 'same' + 1 stride) + Maxpooling (1 pooling size) + 1 Dropout layer (0.001) + Convolutional layer (64 filters + 2 filter size + ELU activation + padding 'same' + 1 stride) + MaxPooling (2 pooling sizes) + 1 Dropout layer (0.001) + 1 LSTM layer (32 neurons + ELU activation) + 1 Dropout layer (0.001) + Flatten + 1 Dense layer (8 neuron, linear activation) + 1 Dropout layer (0.001) + 1 Dense layer (1 neuron) + Compile (MSE loss, Adagrad optimizer with a learning rate of 0.01)

Table 3 The results of the developed DL and ML models for predicting DO and Chl-a in Small Prespa Lake

Variable	Metric	Training period					Testing period				
		LSTM	CNN	CNN–LSTM	SVR	DT	LSTM	CNN	CNN–LSTM	SVR	DT
DO	*r*	0.960	0.960	0.960	0.944	0.949	0.968	0.967	0.970	0.953	0.932
	RMSE (mg/L)	0.680	0.632	0.682	0.734	0.706	0.545	0.658	0.518	0.804	0.764
	MAE (mg/L)	0.521	0.472	0.526	0.529	0.511	0.410	0.522	0.392	0.634	0.569
	PBIAS	− 2.733	− 1.324	− 2.922	0.251	0.001	1.050	3.983	0.221	5.241	0.531
	E_{NS}	0.907	0.920	0.907	0.892	0.900	0.932	0.902	0.939	0.854	0.868
	RRMSE (%)	8.007	7.451	8.030	8.646	8.318	5.668	6.837	5.381	8.354	7.946
	RMAE (%)	8.666	7.125	8.046	7.741	6.751	4.425	5.629	4.176	6.730	6.059
	WI	0.971	0.978	0.972	0.971	0.973	0.984	0.977	0.982	0.965	0.962
Chl-a	*r*	0.924	0.923	0.9237	0.893	0.913	0.869	0.872	0.874	0.854	0.844
	RMSE (mg/L)	0.641	0.651	0.6465	1.151	0.655	0.614	0.611	0.579	0.766	0.646
	MAE (mg/L)	0.325	0.328	0.3087	0.974	0.344	0.433	0.432	0.359	0.618	0.404
	PBIAS	1.828	3.395	4.332	− 23.895	0.005	− 4.617	− 5.194	2.190	− 11.570	− 0.893
	E_{NS}	0.841	0.836	0.839	0.489	0.834	0.727	0.730	0.757	0.575	0.698
	RRMSE (%)	23.447	23.808	23.636	42.092	23.972	24.445	24.325	22.05	30.505	25.705
	RMAE (%)	11.701	11.552	10.197	53.191	11.693	21.507	21.580	15.093	33.534	16.773
	WI	0.940	0.938	0.9416	0.845	0.950	0.877	0.879	0.905	0.742	0.918

The scatter plots of the observed and predicted DO in the testing period (Fig. 6) show that the CNN–LSTM model was more accurate than the other models. It had the highest *r* and was well described by the linear regression equation y = 0.9065x − 0.8789. The DT model had more

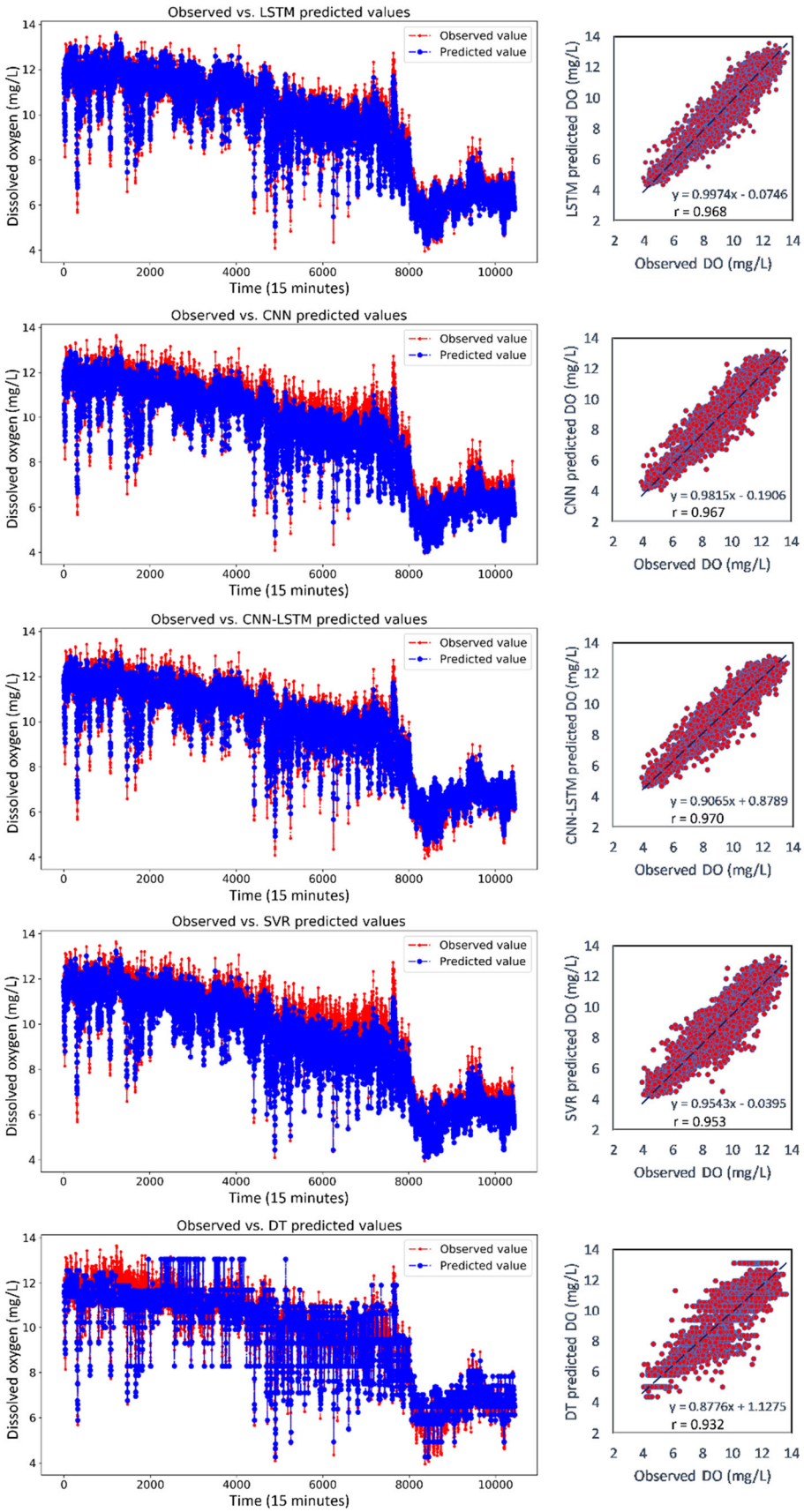


Fig. 6 Plots of the observed and predicted DO concentrations using the models developed

scattered DO predictions, and its linear regression equation was y = 0.8776x − 1.1275.

For Chl-a prediction, the predictive power of the LSTM (r = 0.869, $RMSE$ = 0.614mg/L, MAE = 0.433mg/L, E_{NS} = 0.727, and WI = 0.877) and CNN (r = 0.872, $RMSE$ = 0.611mg/L, MAE = 0.432mg/L, E_{NS} = 0.730, and WI = 0.879) models were similar. The hybrid CNN–LSTM model had an r of 0.874, an $RMSE$ of 0.579 mg/L, an MAE of 0.359 mg/L, an E_{NS} of 0.757 and a WI of 0.905. The $RMAE$ of the CNN–LSTM-based Chl-a prediction model was 6.41% and 6.48% lower than the LSTM and CNN-based Chl-a prediction models in the testing phase, respectively. Therefore, the hybrid CNN–LSTM model outperformed the LSTM and CNN models in predicting Chl-a values. Based on the E_{NS} values for the predictive modeling of Chl-a in the testing phase, the hybrid CNN–LSTM model (E_{NS} = 0.757) had a very good performance, while the other models, i.e., LSTM (0.727), CNN (0.730), SVR (0.575) and DT (0.698), exhibited only good performances. Also, according to the $PBIAS$ metric in the testing period, all models developed for Chl-a prediction had very good performances, except the SVR model, which only had a good performance. Again, the DL models performed better than the traditional ML models. The scatter plots of the observed and predicted Chl-a concentrations in the testing period (Fig. 7) show that the hybrid CNN–LSTM predictions, fitted by the linear regression equation y = 0.6998x − 0.7008, resulted in more accurate and less scattered predictions. In contrast, the DT model was the least accurate and most scattered, with a linear regression equation of y = 0.8121x − 0.4951.

The models' efficiency was investigated using Taylor diagrams (Fig. 8). For DO prediction, the CNN–LSTM, with a correlation coefficient of 0.97 and a normalized standard deviation of 0.93, had the best results. The SVR model, with a correlation coefficient of 0.95 and a normalized standard deviation of 1.0, had the worst results. Similar to DO prediction, the CNN–LSTM-based Chl-a predictive model, with a correlation coefficient of 0.87 and normalized standard deviation of 0.79, was the best at predicting Chl-a. Once again, the SVR model, with a correlation coefficient of 0.85 and normalized standard deviation of 0.57, performed the worst of all the models. In summary, the CNN–LSTM hybrid model was the best at predicting both DO and Chl-a.

Figure 9 shows the box plots of the observed and predicted values of DO and Chl-a in Small Prespa Lake. The median values generated by the LSTM and CNN–LSTM models were similar to the observed DO values. These models differed in the lower quartile, the 25th percentile (Q_{25}), and in data range (maximum and minimum). The Q_{25} for the CNN–LSTM model was similar to the observed DO values, while the data range of the LSTM model was close to the observed DO values. Although the medians for the LSTM, CNN and CNN–LSTM models were close to the observed Chl-a values, the Q_{25} and minimum values were closer to the observed Chl-a values. Also, the upper outliers for the CNN–LSTM model were closest to the observed Chl-a values, followed by the CNN and LSTM models. For both the DO and Chl-a variables, the SVR model had the most different statistical variables, indicating that it performed the worst out of all the models. Overall, the DL-based prediction models closely matched the observed values within the testing period, while the SVR and DT models failed to do so. The DT and SVR models require well-structured data for more accurate predictions (Bui et al. 2020). However, DL models are powerful predictive models even when the data are poorly structured.

Figure 10a, b illustrate the spider-plots of the $RRMSE$ and E_{NS} for the models, which were used to validate the models' performance. For both the DO and Chl-a simulations, the CNN–LSTM yielded the lowest $RRMSE$ and the largest E_{NS}, indicating that it was the best performing model. In contrast, the SVR model obtained the highest $RRMSE$ and smallest value of E_{NS}, indicating it was the worst performing model.

The histogram plots of the models' prediction errors were used to investigate the distribution of prediction errors for the different models in the testing period. The plots were generated using several prediction error intervals with widths of ± 0.1 mg/L and ± 0.1 µg/L for the DO and Chl-a simulations, respectively (Fig. 11). For the predictive DO models, the height of the error bin for the CNN–LSTM model showed the greatest frequency (18.1%) for the smallest error magnitude (0.1 mg/L) compared to the LSTM and CNN models, which had the highest frequencies of 14.7% and 12.8% for the error magnitude of 0.2 mg/L, respectively. Similarly, for the Chl-a predictions, the CNN–LSTM model, with a frequency of 21.6%, obtained the largest frequency for the smallest error (0.1 µg/L) compared to the LSTM (14.8%) and CNN (15.4%) models for the error magnitude of 0.4 µg/L. These histogram plots show that the hybrid CNN–LSTM models are more reliable than the LSTM and CNN models when predicting DO and Chl-a as most of their prediction errors occurred within the lowest magnitude band. Generally, the models developed for DO prediction performed better than those developed for Chl-a prediction. This may be due to the linear and nonlinear relationships between the input and output variables. The Pearson's correlation for Chl-a and the input variables (EC (− 0.31), ORP (0.08), pH (0.11) and water temperature (− 0.13)), and for the DO and the input variables (EC (− 0.29), ORP (0.29), pH (0.49) and water temperature (− 0.72)), indicate that the linear relationship between the inputs and output in the DO model

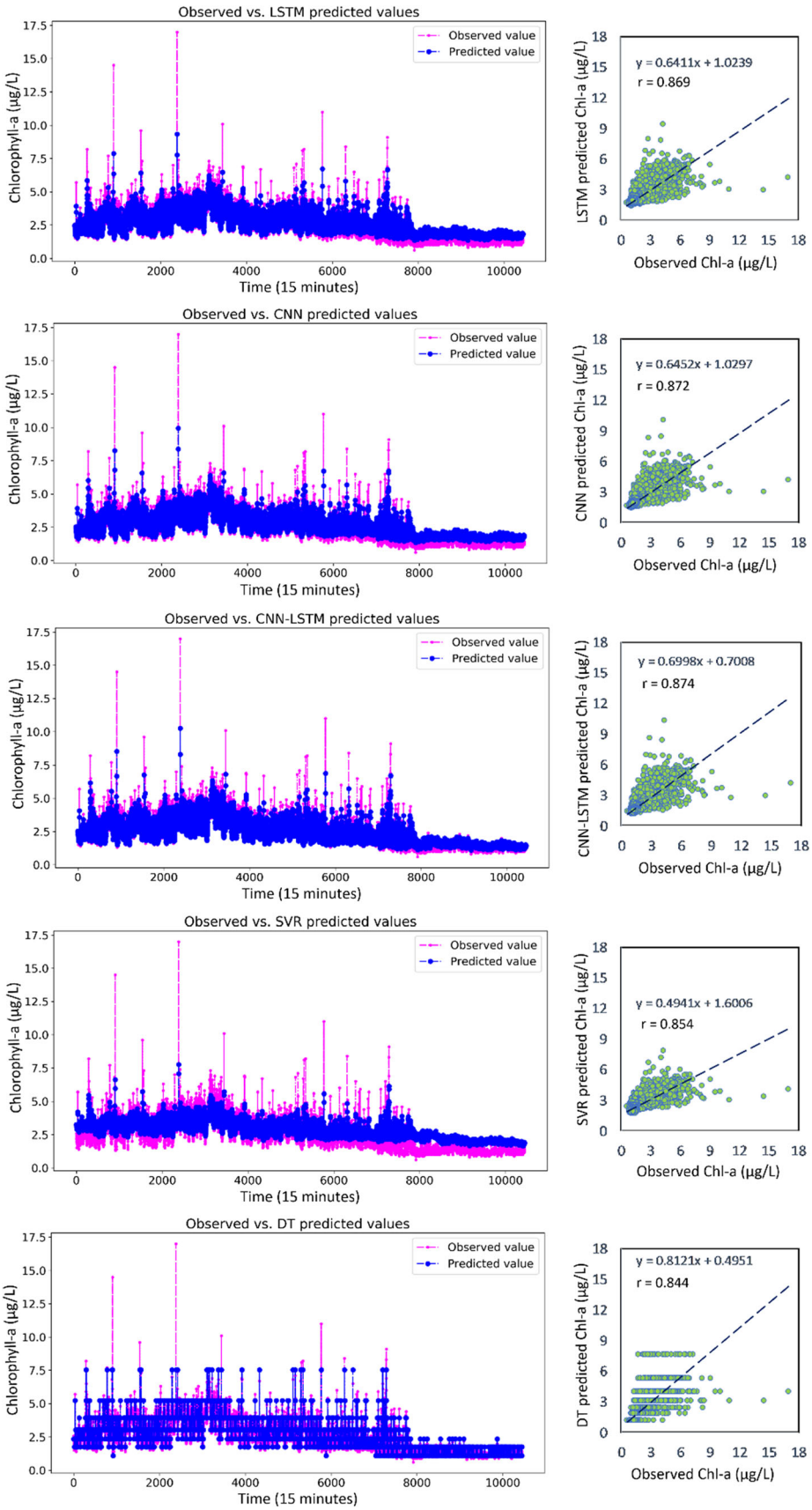


Fig. 7 Plots of the observed and predicted Chl-a concentrations using the models developed

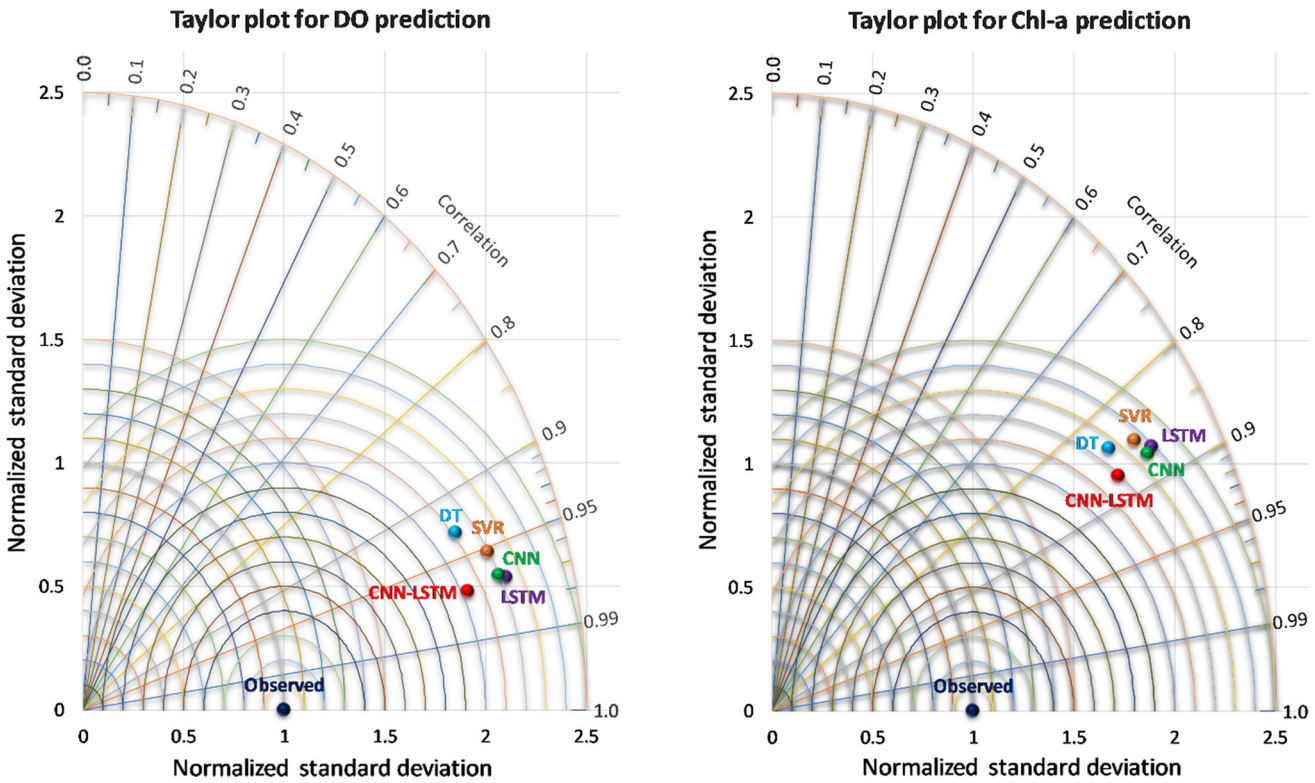


Fig. 8 Taylor plot of the models' performances in predicting DO and Chl- in Small Prespa Lake

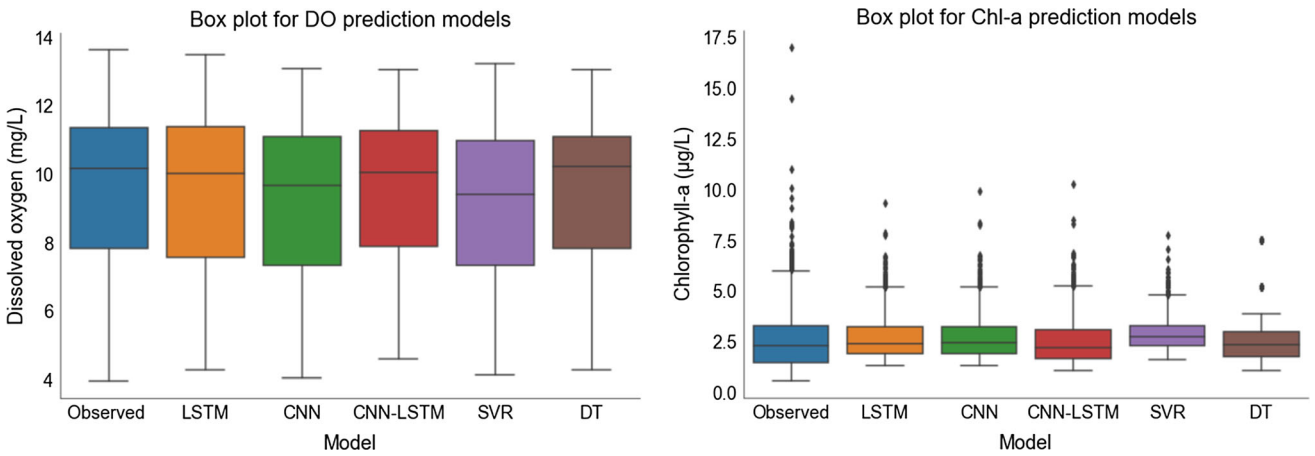


Fig. 9 Box plot comparing the models' abilities to predict DO and Chl-a in Small Prespa Lake

was more significant than the linear relationship between the inputs and output variables in the Chl-a model. Therefore, the prediction of the DO variable is more dependent on the physicochemical variables than the Chl-a variable.

It was observed that all the developed DL models had a high level of learning and predictive power, with minor variations that may be related to their different learning approaches (Bui et al. 2020). Although implementation of the DL models is more complex than traditional ML models, they have higher flexibility in learning nonlinearity, especially in large datasets. Although this study confirmed the capability of the CNN, LSTM and CNN–LSTM models for short-term prediction tasks, Wang et al. (2019) indicated that LSTM is robust with respect to medium-term and long-term simulations. However, as indicated by Li et al. (2017), it is more difficult to make a long-term prediction because long-term prediction tasks require more pertinent historical input data than a short-term prediction. Hence, exploration of the capability of different DL approaches for medium- and long-term water quality predictions is recommended. The hybrid CNN–

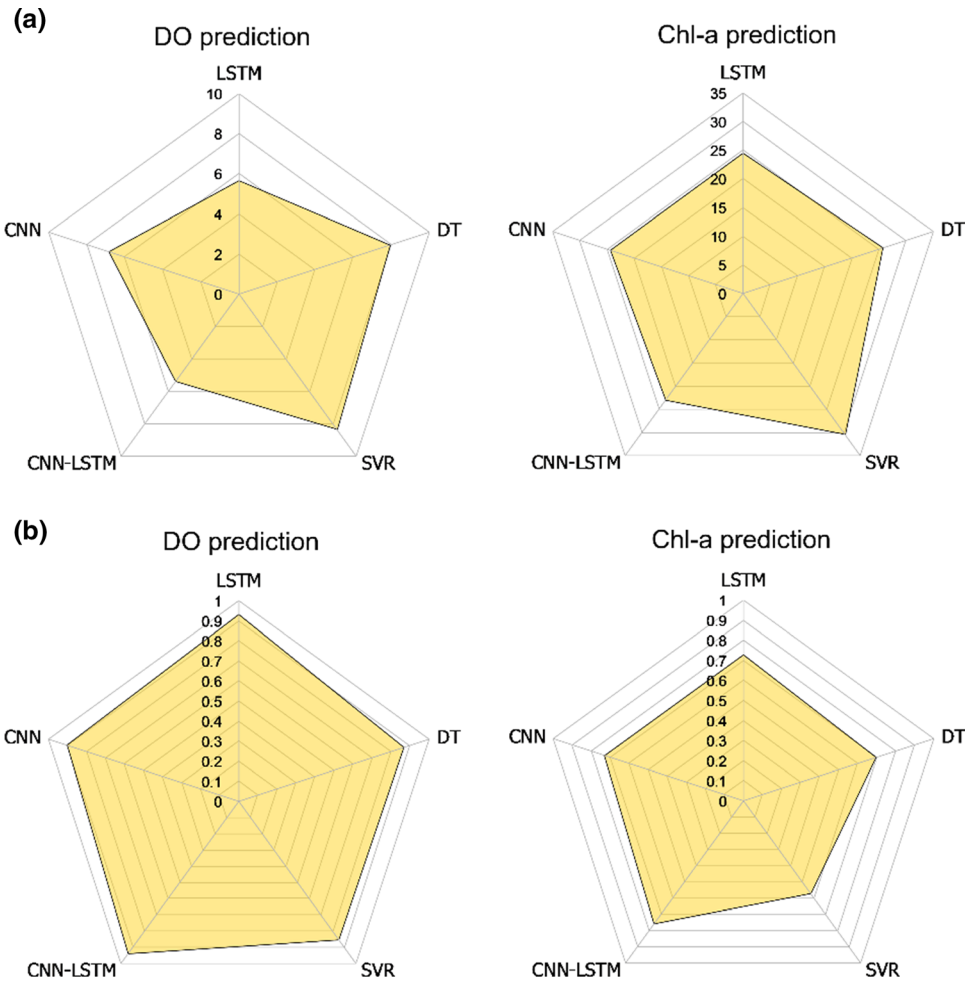


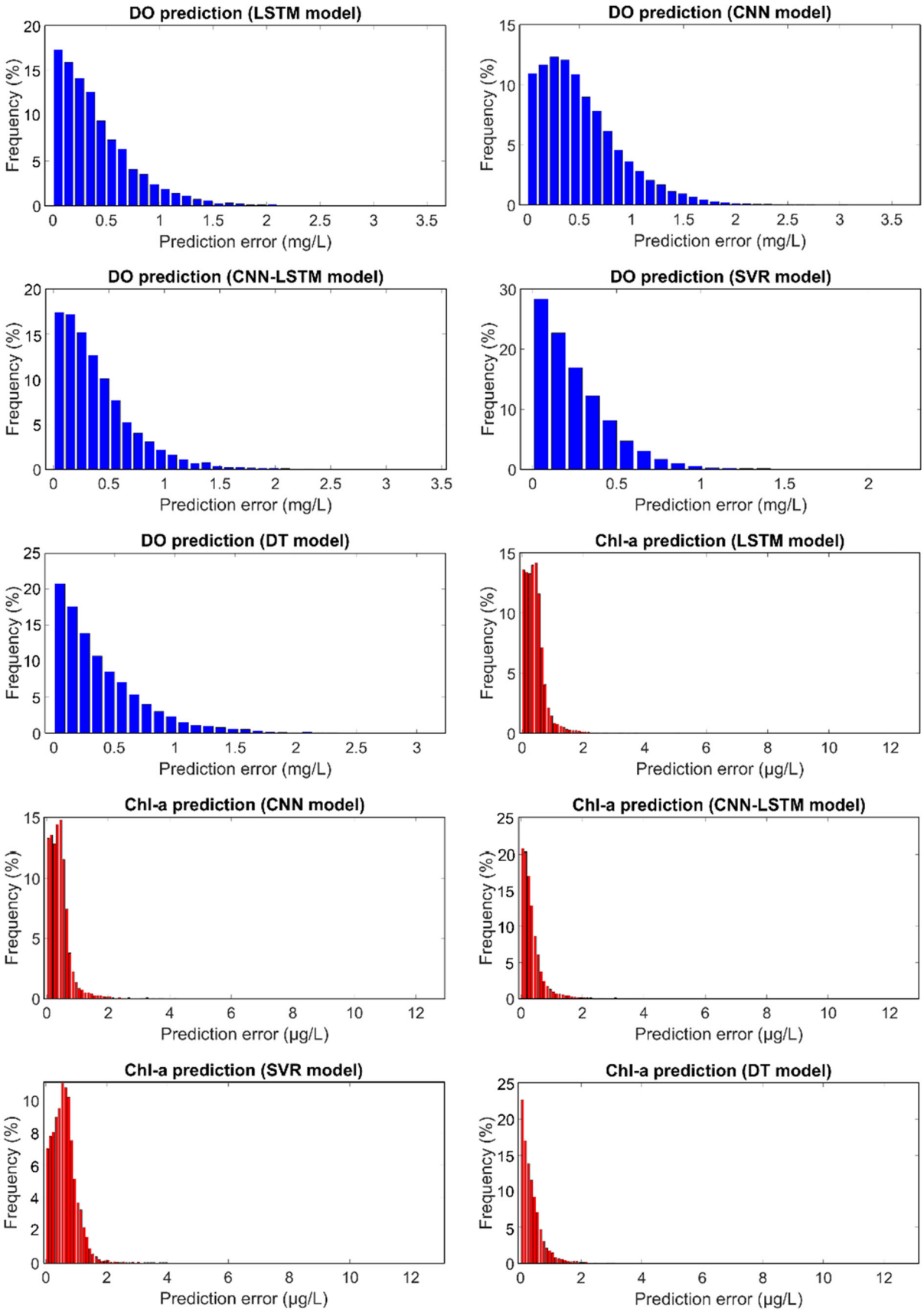
Fig. 10 Spider plot of the models' performances in predicting DO and Chl-a using the developed models in the testing period; **a** relative root mean square error (*RRMSE*; %) and **b** Nash–Sutcliffe model efficiency coefficient (E_{NS})

LSTM model outperformed the standalone individual LSTM and CNN models, as well as the ML models. This is because the LSTM is capable of learning long-term dependency and CNN can extract time-invariant features. Thus, their combination improved the overall accuracy. In conventional ML models, most of the features need to be classified by a domain expert to reduce the complexity of the data and make patterns more visible to the learning algorithms. In contrast, the DL models learn high-level features from data in an incremental manner, which removes the need for domain expertise and feature extraction. The results established that the low-cost model developed in this study, which used easily measured water variables, can be an alternative to monitor water quality in Small Prespa Lake. The proposed model can also be used to rebuild past Chl-a and DO conditions using these same easily measured variables.

Future studies should explore the ability of DL models to simulate water quality parameters over medium- and long-term periods. Preprocessing methods, e.g., wavelet decomposition combined with DL models, could also be explored to predict water quality. Using decomposition methods, some of the noise of the data is removed, and the accuracy of the predictive model can sometimes be improved. The methodology used in this study could also be applied to predict other hydrological variables. Furthermore, other DL models, e.g., stacked auto-encoder, restricted Boltzmann machine deep reinforcement learning, and generative adversarial network, could also be interesting to explore for water quality prediction.

4 Conclusions

This research used four easily measured water quality variables, i.e., EC, pH, ORP, and water temperature, to construct DL (i.e., LSTM, CNN, and hybrid CNN–LSTM for the first time in the field of water quality modeling) and traditional ML (i.e., SVR and DT) models to predict DO and Chl-a concentrations in Small Prespa Lake, Greece. Water quality data were collected at 15-min intervals from June 1, 2012 to May 31, 2013. The input dataset was transformed into a supervised frame and then split into training (70%) and testing (30%) datasets. The models


DO prediction (LSTM model)
DO prediction (CNN model)
DO prediction (CNN-LSTM model)
DO prediction (SVR model)
DO prediction (DT model)
Chl-a prediction (LSTM model)
Chl-a prediction (CNN model)
Chl-a prediction (CNN-LSTM model)
Chl-a prediction (SVR model)
Chl-a prediction (DT model)
Frequency (%)
Prediction error (mg/L)
Prediction error (μg/L)

◀**Fig. 11** The frequency plots of the absolute prediction error produced by the models in the testing period for DO and Chl-a predictions

were trained using the optimal hyper-parameters for each model. The testing period data were used to evaluate the performance of the models using statistical criteria: *r*, *RMSE*, *MAE* and their normalized equivalents expressed as a percentage (*RRMSE, RMAE*), along with E_{NS} and *WI*.

Although the LSTM model outperformed the CNN model in predicting DO concentrations, their performances were similar for Chl-a prediction. The performance evaluation of the hybrid CNN–LSTM model, which has the strengths of both standalone DL models, demonstrated that it decreased the *RMAE* for DO prediction by 0.24% and 1.45% in the testing period compared to the LSTM and CNN models, respectively. For Chl-a prediction, the hybrid CNN–LSTM model decreased the *RMAE* by 6.41% and 6.48% in the testing period compared to the LSTM and CNN models, respectively. The LSTM models were much better at predicting higher levels of DO and Chl-a concentrations in the testing period, while the CNN model was better at predicting the lower levels of these variables. By integrating these models, the hybrid CNN–LSTM model successfully captured both low and high levels of the water quality variables, particularly for the DO concentrations.

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